

EC301 · SVNIT SURAT · THE TU CHAHIYE QUIZ, ON PAPER

DIGITAL COMMUNICATION

40 QUESTIONS

90 MARKS

45 MINUTES

ANSWER KEY AT THE BACK

How the marks work. Ten easy questions at 1 mark, ten medium at 2, and twenty hard at 3 — ninety in total, two thirds of them in the hard twenty. One option is right in every question. **The answer key, with a short explanation for every question, starts on the last section of this document.**

40

QUESTIONS

90

MARKS

45:00

ON THE CLOCK

1

ATTEMPT ONLINE

- 01** What are the three steps of PCM, in order? 1 MARK
- A Encoding, sampling, quantization B Sampling, quantization, encoding
C Quantization, encoding, sampling D Filtering, sampling, coding
- 02** The Nyquist sampling criterion says the sampling frequency must be at least... 1 MARK
- A equal to the highest frequency in the signal B twice the highest frequency in the signal
C half the highest frequency in the signal D ten times the highest frequency in the signal
- 03** In Shannon's sense, which message carries the most information? 1 MARK
- A The longest one B The most important one
C The one about the least likely event D The one that arrives first
- 04** What does AMI stand for, and what is its rule? 1 MARK
- A Alternate Mark Inversion — every 1 flips the sign of the previous 1 B Amplitude Modulated Inversion — every bit inverts the carrier
C Automatic Mark Indication — every 0 is marked D Alternate Multilevel Indexing — levels cycle through four values
- 05** What does ISI stand for? 1 MARK
- A Inter-Symbol Interference B Internal Signal Inversion
C Instantaneous Signal Integration D Inter-Sample Isolation
- 06** In BPSK, what is changed to carry the data? 1 MARK
- A The amplitude of the carrier B The frequency of the carrier
C The phase of the carrier D The bandwidth of the carrier

- 07** Source coding and channel coding do opposite things. Which is which? 1 MARK
- A Source coding adds redundancy; channel coding removes it B Source coding removes redundancy; channel coding adds it
- C Both add redundancy D Both remove redundancy
- 08** What does the 'A' in AWGN stand for, and what does it mean? 1 MARK
- A Analog — the noise is continuous B Additive — the noise simply adds to the signal
- C Attenuated — the noise is weakened by the channel D Amplified — the noise grows with distance
- 09** Entropy $H(X)$ is measured in which units? 1 MARK
- A Bits per second B Bits per symbol
- C Symbols per second D Watts per hertz
- 10** For a fixed capacity, the Shannon–Hartley law says you can trade bandwidth against... 1 MARK
- A signal power B frequency
- C cable length D the number of users
- 11** An 8-bit quantizer is used on a full-scale sine wave. What is the SQNR? 2 MARKS
- A 48.00 dB B 49.92 dB
- C 56.00 dB D 24.08 dB
- 12** Why does an RZ line code need twice the bandwidth of NRZ? 2 MARKS
- A It uses twice as many voltage levels B Its pulse is only half a bit wide, so it changes twice as fast
- C It sends every bit twice D It has a DC component that must be filtered
- 13** What is the roll-off factor α of a raised-cosine pulse, and what does $\alpha = 0$ mean? 2 MARKS
- A The fraction of bandwidth wasted; $\alpha = 0$ means no bandwidth at all B How gently the spectrum tapers; $\alpha = 0$ is the ideal brick-wall filter at the Nyquist minimum
- C The number of zero crossings; $\alpha = 0$ means none D The attenuation in dB; $\alpha = 0$ means lossless
- 14** What is the maximum possible entropy of a source with 8 symbols, and when does it occur? 2 MARKS
- A 8 bits, when one symbol is certain B 3 bits, when all eight are equally likely
- C 1 bit, when the source is binary D Unlimited, it depends on the symbol rate
- 15** Why is a prefix-free code useful? 2 MARKS
- A It uses fewer bits than any other code B No codeword starts another, so the decoder knows a codeword has ended the moment it ends
- C It always satisfies the Kraft inequality with equality D It removes the need for a synchronisation pattern

16 A code has minimum Hamming distance $d(\min) = 5$. How many errors can it detect and how many can it correct? 2 MARKS

- A Detect 5, correct 2
B Detect 4, correct 2
C Detect 4, correct 3
D Detect 2, correct 1

17 What does an eye diagram tell you at a glance? 2 MARKS

- A** The bit error rate exactly
- B** How much ISI and timing jitter there is — a wide open eye means a large margin
- C** The carrier frequency
- D** The number of quantization levels

18 Why is BPSK about 3 dB better than ASK for the same bit error rate? 2 MARKS

- A** It uses twice the bandwidth
- B** Its two symbols sit at $-A$ and $+A$, which is twice as far apart as ASK's 0 and A
- C** It has a constant envelope
- D** Its receiver is non-coherent

19 What is a scrambler for? 2 MARKS

- A** Encrypting the data so it cannot be read
- B** Breaking up long runs of identical bits and periodic patterns so the receiver's clock stays locked
- C** Compressing the data before transmission
- D** Adding parity bits for error detection

20 In a convolutional encoder, what does the constraint length K mean? 2 MARKS

- A The number of output bits per input bit
- B The number of input bits that influence any one output bit
- C The length of the whole message
- D The number of states in the trellis

21 A television signal of bandwidth 4.2 MHz is sent by binary PCM with 512 quantization levels. Find the codeword length and the bit rate. 3 MARKS

- A** $n = 9$ bits, $R_b = 75.6$ Mb/s
- B** $n = 8$ bits, $R_b = 67.2$ Mb/s
- C** $n = 9$ bits, $R_b = 37.8$ Mb/s
- D** $n = 10$ bits, $R_b = 84$ Mb/s

22 A 3.5 kHz signal is sampled, quantized and coded, and the channel can carry only 50 kbit/s. $V_{pp} = 4$ V and $V_{rms} = 0.2$ V. What are n and the step size? 3 MARKS

- A** $n = 8, \Delta = 0.0156$ V **B** $n = 7, \Delta = 0.03125$ V
C $n = 7, \Delta = 0.0625$ V **D** $n = 14, \Delta = 0.000244$ V

23 Continuing that problem — $V_{\text{rms}} = 0.2 \text{ V}$ and $\Delta = 0.03125 \text{ V}$. What is the SNR in dB? 3 MARKS

- A** 49.92 dB **B** 43.94 dB
C 26.91 dB **D** 13.46 dB

- 24** A channel of bandwidth 75 kHz carries binary data at 0.1 Mb/s using raised-cosine pulses. What is the roll-off factor? 3 MARKS
- A $\alpha = 0.25$ B $\alpha = 0.5$
C $\alpha = 0.75$ D $\alpha = 1.0$
- 25** Eight 2 kHz signals are time-division multiplexed using binary PCM. The sampling error must not exceed 1% of the peak amplitude, and the sampling rate must be at least 25% above Nyquist. Find the aggregate bit rate. 3 MARKS
- A 224 kb/s B 280 kb/s
C 320 kb/s D 140 kb/s
- 26** A telephone line has a bandwidth of 3 kHz and an SNR of 30 dB. What is its Shannon capacity? 3 MARKS
- A About 90 kb/s B About 30 kb/s
C About 9 kb/s D About 3 kb/s
- 27** A link has $B = 1$ MHz, a noise density of -174 dBm/Hz and a received signal power of -100 dBm. What is the SNR in dB? 3 MARKS
- $B = 1 \times 10^6$ Hz
 $N_0 = -174$ dBm/Hz
 $S = -100$ dBm
- A 14.0 dB B 74.0 dB
C -274 dB D 60.0 dB
- 28** A binary symmetric channel has a crossover probability $p = 0.1$. What is its capacity per symbol? 3 MARKS
- A 0.9 bits/symbol B 0.531 bits/symbol
C 0.469 bits/symbol D 0.1 bits/symbol
- 29** Does a binary prefix-free code with codeword lengths 1, 2, 2, 3 exist? 3 MARKS
- A Yes, and it is perfectly efficient B Yes, but only with a fifth codeword added
C No, because the Kraft sum comes to 1.125 D No, because prefix-free codes must all be the same length
- 30** Build a Shannon–Fano code for probabilities 0.35, 0.20, 0.15, 0.15, 0.10, 0.05. What is the average codeword length? 3 MARKS
- A 0.35 B 0.20 C 0.15
D 0.15 E 0.10 F 0.05
- A $L = 2.36$ bits/symbol B $L = 2.45$ bits/symbol
C $L = 2.58$ bits/symbol D $L = 3.00$ bits/symbol

- 31** Two students build different Huffman codes for the same source. Can both be correct? 3 MARKS
- A No, the Huffman code is unique
B Yes — the 0/1 labelling and tie-breaking are free choices, but the average length L must come out the same
C Yes, and the average lengths may differ too
D Only if they use the same number of symbols
- 32** A (7,4) linear block code is used. What is its code rate, and what does a lower code rate mean? 3 MARKS
- A $7/4 = 1.75$; lower means less redundancy
B $4/7 \approx 0.571$; lower means more redundancy and better protection
C $4/7 \approx 0.571$; lower means less protection
D $3/7 \approx 0.429$; lower means faster transmission
- 33** Compute the CRC for the message 11001011 with the divisor 1011, and give the transmitted codeword. 3 MARKS
- message : 11001011
divisor : 1011
- A CRC 111, codeword 11001011111
B CRC 011, codeword 11001011011
C CRC 101, codeword 11001011101
D CRC 110, codeword 11001011110
- 34** A rate-1/2, $K = 3$ convolutional encoder has $g_1 = 111$ and $g_2 = 101$. What does the message 10011 produce? 3 MARKS
- $g_1 = 111 \rightarrow c_1 = \text{input XOR FF1 XOR FF2}$
 $g_2 = 101 \rightarrow c_2 = \text{input XOR FF2}$
message = 1 0 0 1 1
- A 11 10 11 11 01 01 11
B 11 01 10 01 11 00 11
C 10 11 01 11 10 01 11
D 11 11 10 10 01 01 11
- 35** Why does every practical Viterbi decoder use a trellis rather than a tree? 3 MARKS
- A The tree cannot represent all the paths
B The tree doubles in size at every step, while the trellis folds the repeated states so the work per step is constant
C The trellis is easier to draw by hand
D The tree only works for rate-1/2 codes
- 36** For a BPSK modulator with a carrier at 70 MHz and a bit rate of 10 Mbps, what are the side frequencies and the minimum bandwidth? 3 MARKS
- A 65 and 75 MHz; bandwidth 10 MHz
B 60 and 80 MHz; bandwidth 20 MHz
C 69.5 and 70.5 MHz; bandwidth 1 MHz
D 70 and 80 MHz; bandwidth 10 MHz
- 37** An FSK modulator has $f_1 - f_2 = 100$ kHz and a data rate of 1 Mbit/s with full-width pulses. What is the transmission bandwidth? 3 MARKS
- A 1.1 MHz
B 2.0 MHz
C 2.1 MHz
D 100 kHz

- 38** Which of these line codes suppresses the DC component, and why does that matter? 3 MARKS
- A Unipolar NRZ, because it only uses positive voltages B Bipolar AMI and Manchester, because their average voltage is zero
- C Polar NRZ, because it uses two symmetric levels D None of them — all line codes have a DC component
- 39** In the three-tap zero-forcing equalizer designed in your tutorial, what condition do the tap weights enforce? 3 MARKS
- A That the output pulse is maximised at every sampling instant B That $p(0) = 1$ while $p(-1) = p(1) = 0$, so the neighbouring samples contribute nothing
- C That the noise power is minimised D That the channel response becomes flat at all frequencies
- 40** A duobinary receiver reads a long run of sample values including several 0s and ± 2 s. What tells it that a detection error has occurred? 3 MARKS
- A Any sample value of 0 B Two non-zero samples of the same polarity separated only by zeros
- C Any sample value of ± 2 D A sample value outside the range -2 to $+2$

The answer key

The letter, then why. Read the explanation even where you got it right — that is where most of the revision value is.

- 01 **B** Sampling measures the signal's height at regular times, quantization rounds each measurement to the nearest allowed level, and encoding writes each level as a binary number. Sampling digitises the time axis; quantization digitises the amplitude axis.
- 02 **B** $f_s \geq 2 \cdot f_m$. You need at least two samples per cycle of the fastest wiggle — one for the peak, one for the trough. Fewer than two and a fast wave can masquerade as a slow one, which is aliasing, and it can never be undone afterwards.
- 03 **C** Information depends only on the probability of the event, not on its content, meaning or importance. A message about the least likely event carries the most information — it is a measurement of surprise.
- 04 **A** Alternate Mark Inversion. 'Mark' is old telegraph language for a 1, so every 1 takes the opposite polarity to the 1 before it: +A, -A, +A ... while 0s sit at the zero level. Because the 1s alternate, the average voltage is zero, which removes the DC component.
- 05 **A** Inter-Symbol Interference. A bandwidth-limited channel smears each pulse in time, so the tail of one pulse is still present when its neighbour is sampled, and the receiver reads the wrong value. It is the echo in a large hall, in electrical form.
- 06 **C** The phase. A 1 sends $A \cdot \cos(2\pi f_c \cdot t)$ and a 0 sends the same wave shifted by 180 degrees, which is $-A \cdot \cos(2\pi f_c \cdot t)$. The amplitude and the frequency stay exactly the same throughout — so BPSK is really just polar NRZ multiplied by a carrier.
- 07 **B** Source coding squeezes the data by removing the useless redundancy that was already there by accident. Channel coding then adds back a small amount of carefully designed redundancy built to fight noise. You throw out the junk, then add the right tool.
- 08 **B** Additive: $y(t) = x(t) + n(t)$. The noise adds to the signal rather than multiplying or twisting it. White means flat across all frequencies, Gaussian means the voltage follows the bell curve, and Noise means unwanted random energy, mostly thermal.
- 09 **B** Bits per symbol — it is the average information per symbol. Multiply it by the symbol rate r to get the information rate $R = r \cdot H(X)$, which is in bits per second. Mixing the two up is the most common slip in this topic.
- 10 **A** $C = B \cdot \log_2(1 + S/N)$. Bandwidth sits outside the logarithm and power sits inside it, so you can hold C constant by spreading the signal thinly over a lot of spectrum and whispering, or squeezing into a narrow band and shouting. GPS does the first; a broadcast tower does the second.
- 11 **B** $SQNR(\text{dB}) = 6.02n + 1.76 = 6.02 \times 8 + 1.76 = 48.16 + 1.76 = 49.92 \text{ dB}$. Remember the headline it contains: every extra bit buys about 6 dB. But use this formula only when the signal is a full-scale sine wave — if a V_{rms} is given, use $V_{\text{rms}}^2 / (\Delta^2/12)$ instead.
- 12 **B** Return-to-Zero puts the pulse in the first half of the bit and returns to zero for the second half. A pulse half as wide in time contains twice the frequency content, and bandwidth is roughly the inverse of pulse width. The payoff is a guaranteed transition in every 1-bit, which helps the receiver's clock.
- 13 **B** $BT = (R_b/2)(1 + \alpha)$. $R_b/2$ is the Nyquist minimum, and $(1 + \alpha)$ is the penalty above it. $\alpha = 0$ is the ideal brick wall — perfect on paper, impossible to build, and its tails decay so slowly that a tiny timing error causes huge ISI. $\alpha = 1$ doubles the bandwidth but is very forgiving. Real systems use 0.2 to 0.5.
- 14 **B** $0 \leq H(X) \leq \log_2(m)$. With $m = 8$, $\log_2(8) = 3$ bits per symbol, reached when all eight symbols are equally likely — maximum unpredictability means maximum average information. The minimum, 0, happens when one symbol has probability 1.

- 15 **B** Prefix-free means no codeword is the beginning of another one. The instant a valid codeword has been received you know it is complete, so decoding is instantaneous — bit by bit, no waiting and no backtracking. Prefix-free is sufficient for unique decodability, but not necessary: some non-prefix codes are still uniquely decodable, just not instantaneous.
- 16 **B** Detect up to $d(\min) - 1 = 4$, correct up to $\text{floor}((d(\min) - 1)/2) = 2$. Detection works because changing up to 4 bits can never turn one valid codeword into another. Correction works by choosing the nearest valid codeword, and with 2 errors you are still closer to the original (2 away) than to any other (3 away).
- 17 **B** It is many bit periods of the received waveform drawn on top of each other. The widest vertical opening is the noise margin and the best sampling instant; the width of the crossings at the bottom is the timing jitter. ISI and noise close the eye, so an eye that is open is a system that is working.
- 18 **B** P_e for ASK and FSK is $Q(\sqrt{E_b/N_0})$; for BPSK it is $Q(\sqrt{2 \cdot E_b/N_0})$. That factor of two is 3 dB, and geometrically it is just the distance between the symbols: ASK puts them A apart, BPSK puts them $2A$ apart. Whenever you can, spend power moving symbols apart rather than turning them off.
- 19 **B** It is not encryption — the descrambler is public and anybody can undo it. Its job is to remove long strings of 0s and repetitive patterns from the bit stream, because those give the receiver nothing to lock its clock onto and can also put a large DC component on the line.
- 20 **B** K is how many input bits affect any single output bit — the current one plus the $K-1$ held in the shift registers. For the rate-1/2, $K = 3$ encoder there are $K - 1 = 2$ flip-flops, so $2^2 = 4$ states in the trellis, and the generator vectors $g_1 = 111$ and $g_2 = 101$ say which taps each adder is connected to.
- 21 **A** $f_s = 2 \cdot f_m = 8.4$ MSamples/s. $n = \log_2(512) = 9$ bits per sample, since $512 = 2^9$. $R_b = n \cdot f_s = 9 \times 8.4 \times 10^6 = 75.6$ Mb/s. For completeness the SQNR is $6.02(9) + 1.76 = 55.94$ dB, and the polar-NRZ transmission bandwidth is $R_b/2 = 37.8$ MHz.
- 22 **B** $f_s = 2 \times 3.5$ kHz = 7 kSamples/s. $n(\max) = R_b/f_s = 50,000/7,000 = 7.14$, and since n must be a whole number and the channel rate must not be exceeded, round DOWN to $n = 7$. Then $\Delta = V_{pp}/2^n = 4/128 = 0.03125$ V. (Round down for a channel-rate limit; round up for an accuracy requirement.)
- 23 **C** A_{Vrms} is given, so use the general form, not $6.02n + 1.76$. $SNR = V_{rms}^2 / (\Delta^2/12) = 0.04 / (0.0009766/12) = 0.04 / 0.00008138 = 491.5$. In dB: $10 \cdot \log_{10}(491.5) = 26.91$ dB. It is far below the 43.9 dB a full-scale sine would give at 7 bits, because this signal is quiet — 0.2 V rms against a possible 1.414 V rms.
- 24 **B** $T_b = 1/R_b = 10^{-5}$ s. From $f_B = (1 + \alpha)/(2T_b)$, rearrange: $1 + \alpha = 2 \cdot f_B \cdot T_b = 2 \times 75,000 \times 10^{-5} = 1.5$, so $\alpha = 0.5$. Sanity check: the Nyquist minimum for 0.1 Mb/s is 50 kHz, and 75 kHz is 1.5 times that.
- 25 **B** Nyquist rate = 2×2 kHz = 4 kHz; 25% above gives $f_s = 5$ kHz per channel. The error rule: $\Delta/2 \leq 0.01A$, and with $\Delta = 2A/L$ this gives $L \geq 100$, so the next power of two is 128 and $n = 7$ bits. Aggregate = 8 channels $\times$ 5000 samples/s $\times$ 7 bits = 280 kb/s. With raised-cosine pulses at $\alpha = 0.2$ the bandwidth would be $(280k/2)(1.2) = 168$ kHz.
- 26 **B** First convert the dB: $30 = 10 \cdot \log_{10}(S/N)$ so $S/N = 10^3 = 1000$. Then $C = B \cdot \log_2(1 + S/N) = 3000 \times \log_2(1001) = 3000 \times 9.97 \approx 29,900$ bit/s. Never put a dB value straight into the formula — it wants the plain ratio. This number is why dial-up modems topped out near 33.6 kbit/s: they had hit a law of physics.
- 27 **A** $N(\text{dBm}) = N_0 + 10 \cdot \log_{10}(B) = -174 + 10 \cdot \log_{10}(10^6) = -174 + 60 = -114$ dBm. Because both powers are already in dBm, the ratio becomes a subtraction: $SNR = -100 - (-114) = 14.0$ dB. That is $S/N = 25.1$ as a ratio, so $C = 10^6 \times \log_2(26.1) \approx 4.71$ Mb/s.
- 28 **B** $C_s = 1 + p \cdot \log_2(p) + (1-p) \cdot \log_2(1-p) = 1 + 0.1(-3.322) + 0.9(-0.152) = 1 - 0.3322 - 0.1368 = 0.531$ bits/symbol. Read the shape of the formula: capacity is 1 minus what the noise steals. A 10% error rate costs nearly half the capacity — which is why $p = 0.5$ gives zero.
- 29 **C** Kraft: $K = 2^{-1} + 2^{-2} + 2^{-2} + 2^{-3} = 0.5 + 0.25 + 0.25 + 0.125 = 1.125$. Since $K > 1$ the budget is overspent and no instantaneous code with those lengths can exist — the codewords would be forced to collide. Short codewords are expensive: a 1-bit code alone uses half the budget.

- 30 **B** Split into two groups of nearly equal probability at each stage: $\{A,B\} = 0.55$ against $\{C,D,E,F\} = 0.45$. Working through gives $A = 00$, $B = 01$, $C = 100$, $D = 101$, $E = 110$, $F = 111$ — two 2-bit codes and four 3-bit ones. $L = 0.35(2) + 0.20(2) + 0.15(3) + 0.15(3) + 0.10(3) + 0.05(3) = 0.70 + 0.40 + 0.45 + 0.45 + 0.30 + 0.15 = 2.45$ bits/symbol. With $H(X) = 2.364$, the efficiency is $2.364/2.45 = 96.5\%$.
- 31 **B** Three things are free: which branch you call 0 and which 1, how you break ties between equal probabilities, and where a merged probability is placed among equal values when re-sorting. What is NOT free is L — every correct Huffman code for a source has the same average length, because Huffman is provably optimal. So if your L matches, your answer is right even if the codewords differ.
- 32 **B** $R = k/n = 4/7 \approx 0.571$, so about 57% of what goes on the wire is your data and the rest is protection. A code rate is a fraction of useful bits, so it must always lie between 0 and 1 — a value above 1 would mean getting out more information than you put in. Lower rate means more redundancy, better error protection and lower useful throughput.
- 33 **A** The divisor is 4 bits so $r = 3$: append three zeros to get 11001011000 and divide modulo 2, which means XOR at every step and no borrowing. Working down gives a remainder of 111. Replace the three appended zeros with it: the codeword is 11001011111. At the receiver, dividing the whole codeword by 1011 must give a remainder of 000.
- 34 **A** As polynomials, $m(D) = 1 + D^3 + D^4$. Then $c1(D) = (1 + D + D^2)m(D) = 1 + D + D^2 + D^3 + D^6$, so path 1 is 1111001; and $c2(D) = (1 + D^2)m(D) = 1 + D^2 + D^3 + D^4 + D^5 + D^6$, so path 2 is 1011111. Interleaving one bit from each gives (11, 10, 11, 01, 01, 11). The same answer falls out of a clock-by-clock table of the flip-flop contents — do it both ways to check yourself.
- 35 **B** A tree has 2^N branches after N bits — over a million after 20 bits. The trellis notices that the encoder keeps returning to the same small set of states, so it draws one row per state repeated across time. Combined with keeping only the best surviving path into each state, the work per time step stays constant no matter how long the message is.
- 36 **A** The fastest pattern the data can make is 1010..., one full cycle every two bits, so the highest modulating frequency is $f_a = R_b/2 = 5$ MHz. Multiplying two sinusoids gives the sum and the difference: $70 + 5 = 75$ MHz and $70 - 5 = 65$ MHz. Minimum Nyquist bandwidth = $75 - 65 = 10$ MHz, which equals the bit rate.
- 37 **C** $BT = |f_1 - f_2| + 2R_b = 100 \text{ kHz} + 2(1 \text{ MHz}) = 2.1 \text{ MHz}$. Compare PSK at the same data rate: 2 MHz. FSK always needs more bandwidth than PSK, by exactly the frequency separation between the two tones — and that is FSK's main disadvantage.
- 38 **B** AMI alternates the polarity of successive 1s, and Manchester puts a transition in the middle of every bit, so in both cases the long-run average is zero. It matters because transformers and coupling capacitors in the path block steady voltages, so a code with a DC component droops away. Polar NRZ is only DC-free if the 1s and 0s happen to balance — a long run of 1s holds the line at $+A$.
- 39 **B** The taps are chosen so that the equalised pulse has value 1 at its own sampling instant and exactly 0 at the neighbouring ones — that is, the ISI at those instants is forced to zero, which is where the name comes from. For the tutorial's pulse the solution was $c(-1) = 0.210$, $c_0 = 1.13$, $c_1 = 0.318$. Its weakness is that forcing the zeros can amplify noise, which is why minimum-mean-square equalizers exist.
- 40 **B** Duobinary deliberately creates controlled ISI, and the result is a built-in rule: successive non-zero samples must alternate in polarity when separated only by zeros. If two +2s appear in a row with only zeros between them, an error has happened somewhere in between. That free error detection is one of duobinary's attractions, alongside halving the required bandwidth.